

CS 3563: Introduction to DBMS II

Homework 1: Relational Languages

Due Date and Time: 26 Jan. 2026, 11:59 PM IST

Instruction

Total Marks: 30

1. The homework must be completed individually.
2. Referring to external resources beyond the class notes and reference textbooks is not permitted.
3. The submission must be a single PDF and submitted via Moodle.
4. Late submissions will be handled in accordance with the course policy.

Question 1

(3 Marks)

A mailing application, a mess bill statistics application, and a performance analysis application are running on the IITH student database management system. The physical layout of student records is as follows:

Roll No.	Name	Address	GPA	Hostel Name	Mess Bill
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To improve operational efficiency, the structure of student records is changed such that all numeric fields appear in the beginning and all character fields appear in the end. Explain how each of the above applications would have to be altered to operate correctly under the new format.

Question 2

(3 Marks)

In what ways do the relations (in accordance with the relational model) represented by the following tables differ?

Name	Year	Revenue
Bahubali	2015	600
Pushpa	2021	350
Dhurandhar	2025	1000

Year	Name	Revenue
2015	Bahubali	600
2021	Pushpa	350
2025	Dhurandhar	1000
2015	Bahubali	600

Question 3

(6 Marks)

Consider the following SQL query on a student database:

```
select T.cname from Course T, Course S where  
T.credits > S.credits and S.dept = "CSE"
```

Write an efficient relational algebra expression, using only fundamental operators, that is equivalent to this query. Justify your choice.

Question 4**(8 Marks)**

The left outer join, denoted by the symbol LOJ , for relations R and S is defined to be the union of (a) the natural join of R and S , with (b) the set of tuples from R that do not join with any tuples in S , each concatenated with a “null” S tuple (i.e. an S tuple composed of all null fields).

1. Is there a relational algebra equivalent for LOJ , in terms of fundamental operators?
2. Is LOJ associative?
3. Is LOJ commutative?
4. What is the minimum cardinality of $R \text{ LOJ } S$?

Explain your answer.

Question 5**(10 Marks)**

The Employment Bureau of the Labour Ministry maintains a database with the following schema:

COMPANY (cname, type) PROJECT (pname, cname, pstate)
WORKER (EBno, wname, wstate) EXPERIENCE (EBno, cname, years)
INTERVIEW (cname, EBno, date) HIRE (cname, pname, EBno, salary)

The HIRE relation contains information about the current employees hired by companies, whereas the EXPERIENCE relation stores information about the past employment of employees. Is it possible to write an SQL expression to answer the following query: Identify, in lexicographically sorted order, the names of private “fully-centralized” (all projects in a single state) companies and also private “fully-distributed” (maximum of one project per state) companies that, for their projects, hire only local workers who have been previously employed in exactly three companies? If yes, give the expression. If no, explain why.